

Networking Assignment – Subnetting

1. An organization has been assigned the IPv4 network **172.16.20.0/24**. The network administrator needs to divide the network into **4 equal-sized subnets** for four departments. Determine:

- (a) How many bits need to be borrowed?
- (b) New subnet mask in CIDR and dotted-decimal notation.
- (c) Network Address, First Usable IP, Last Usable IP, and Broadcast Address of the **third subnet**.

2. A bank has been allocated the network block **192.168.50.0/24**. The IT department wants to divide this network into **8 equal-sized subnets**. Calculate:

- (a) Number of borrowed bits
- (b) New subnet mask
- (c) Number of usable hosts per subnet
- (d) Network Address and Broadcast Address of the **fifth subnet**.

3. Given the IP address **192.168.100.25/24**, determine:

- (a) Original subnet mask
- (b) If the network is divided into **4 equal subnets**, what will be the new subnet mask?
- (c) Total number of subnets
- (d) Usable hosts per subnet
- (e) Network Address, First Usable IP, Last Usable IP, and Broadcast Address of the subnet containing **192.168.100.25**.

4. An organization has the IPv4 network 10.10.20.0/24 and needs to create 16 equal-sized subnets. Show your calculation and determine:

- (a) Number of bits borrowed
- (b) New CIDR and dotted-decimal subnet mask
- (c) Number of usable hosts per subnet
- (d) Network Address and Broadcast Address of the 10th subnet.

5. An organization is assigned the IPv4 network block **172.16.10.0/24** and needs to create three subnets with the following requirements:

- Subnet A: 100 hosts
- Subnet B: 50 hosts
- Subnet C: 20 hosts

Allocate the subnets sequentially, starting with the largest requirement. State only the **Network Address with CIDR mask** and **Broadcast Address** for each subnet.

5.1 A bank has been allocated the IPv4 block **192.168.20.0/24** and requires four subnets:

- Head Office: 110 hosts
- Branch A: 50 hosts
- Branch B: 25 hosts
- Management: 10 hosts

Using **VLSM**, allocate the address space efficiently, starting with the largest requirement. Determine the **Network Address/CIDR** and **Broadcast Address** for each subnet.

5.2 An organization has the network block **10.10.0.0/23** and needs the following subnets:

- Data Center: 200 hosts
- Administration: 100 hosts
- Accounts: 50 hosts
- IT Support: 25 hosts

Using VLSM, allocate the addresses sequentially from the largest requirement to the smallest. For each subnet, determine only the **Network Address with CIDR mask** and **Broadcast Address**.

5.3 A university is assigned **172.20.0.0/22** and needs to create the following networks:

- Computer Lab: 500 hosts
- Faculty: 200 hosts
- Administration: 100 hosts
- Library: 50 hosts
- CCTV: 25 hosts

Apply VLSM and allocate the address space efficiently. Determine the **Network Address/CIDR** and **Broadcast Address** of each subnet.

5.4 An enterprise has been assigned **10.20.0.0/21**. It requires the following networks:

- Corporate Network: 1000 hosts
- Data Center: 500 hosts
- Branch Network: 250 hosts
- Voice Network: 100 hosts
- Security/CCTV: 50 hosts

Using VLSM, allocate the address space efficiently. State the **Network Address/CIDR** and **Broadcast Address** for each subnet.

5.5 A large enterprise is assigned the IPv4 network block **172.30.0.0/20**. It requires the following networks:

- Data Center: 1500 hosts
- Corporate Users: 1000 hosts
- Branch Offices: 500 hosts
- Voice Network: 250 hosts
- CCTV Network: 100 hosts
- Network Management: 50 hosts
- Guest Network: 25 hosts

Using **VLSM**, allocate the subnets starting from the largest host requirement. Determine only the **Network Address with CIDR mask** and **Broadcast Address** for each subnet.

6. An organization has the IP address **172.30.45.200/20** and wants to divide its original network into **4 equal-sized subnets**. Determine:

- (a) Original Network Address
- (b) Number of borrowed bits
- (c) New Subnet Mask
- (d) Total number of subnets
- (e) Usable hosts per subnet
- (f) Network Address of the subnet containing the given IP
- (g) First Usable IP
- (h) Last Usable IP
- (i) Broadcast Address

7. A bank has been assigned **192.168.40.0/24** and wants to create **8 equal subnets** for different branches. Determine the following for the **second subnet**:

- (a) Network Address
- (b) First Usable IP
- (c) Last Usable IP
- (d) Broadcast Address

Also state the new subnet mask and number of usable hosts per subnet.

8. An organization is given the IPv4 block **14.24.74.0/24**. It needs two subnets where **Subnet A requires 120 addresses** and **Subnet B requires 60 addresses**. Allocate the addresses sequentially, starting with the larger requirement, and determine:

- (a) Network Address with CIDR mask for each subnet
- (b) Broadcast Address for each subnet
- (c) Number of usable host addresses available in each subnet.

9. Given the IP address **10.25.70.100/10**, divide the original network into **8 equal-sized subnets**. Determine:

- (a) Original Network Address
- (b) Number of borrowed bits
- (c) New Subnet Mask in CIDR and dotted-decimal notation
- (d) Total number of subnets
- (e) Usable hosts per subnet
- (f) Network Address of the subnet containing the given IP
- (g) First Usable IP
- (h) Last Usable IP
- (i) Broadcast Address

10. Given the IP address **172.20.35.140/18**, the network administrator wants to divide the original network into **4 equal-sized subnets**. Determine:

- (a) Original Network Address
- (b) Number of bits to be borrowed
- (c) New Subnet Mask in CIDR and dotted-decimal notation
- (d) Total number of subnets
- (e) Usable hosts per subnet
- (f) Network Address of the subnet containing the given IP
- (g) First Usable IP
- (h) Last Usable IP
- (i) Broadcast Address